

1. APPENDIX CONTENTS AND INSTRUCTIONS:

1.1 Solution Designs and Approvals

The guiding framework for IT investments within the Government of Saskatchewan (GoS) is based on providing a One Government, citizen centric approach towards program and service delivery. Achieving this vision requires a consistent approach for application authentication and access, increased re-use of common services, common platforms, integrated platforms and common data. As new solutions are introduced into the Government of Saskatchewan's IT portfolio, considerations for using existing enterprise services must be taken into account. Conversely, developing solutions with re-usability in mind is also valued.

With the vision of a One Government citizen centric approach, and for all solutions intended to house GoS Data, Proponents are required to incorporate design approval stage(s) into their project plans associated with their solution proposals. The iterative design and approval processes include the initial high level solution proposal, and subsequent elaboration within the project for conceptual, logical and physical designs. Design approvals are the responsibility of, and are required by, the Executive Director of Strategy and Architecture, Information Technology Division, Ministry of SaskBuilds and Procurement. GoS reserves the right to request modifications to solution design to meet required GoS standards and the One Government objectives.

To assist Proponents in understanding expectations regarding solutions intended to manage Government of Saskatchewan data, the Appendix is divided into the following supporting sections:

Section 2 "GOVERNMENT OF SASKATCHEWAN ENTERPRISE SERVICE REQUIREMENTS" **applies to all proposals** regardless of where the solution will be hosted (Government provided infrastructure or alternative vendor provided cloud computing environments). This section outlines key Government Enterprise Services and standards that must be incorporated as part of the solution. It also includes additional Enterprise Services which may be considered and leveraged as part of the solution.

Section 3 "CLOUD COMPUTING ENVIRONMENT CONSIDERATIONS" **applies to proposals in which the hosting of the solution will be outside of the Government of Saskatchewan's provided computing environment** (which currently includes a managed "on-premises" data center and private cloud Infrastructure as a Service provider). This section applies to Proponent provided Software as a Service (SaaS) or Managed Services solutions.

Section 4 "GOVERNMENT OF SASKATCHEWAN COMPUTING ENVIRONMENT CONSIDERATIONS" **applies to proposals which are intended to be hosted within the existing Government provided infrastructure.** The section outlines items to be considered if proposing solutions that will be dependent on Government of Saskatchewan's services associated with its managed on-premises data center and/or private cloud services environment.

2. GOVERNMENT OF SASKATCHEWAN ENTERPRISE SERVICE REQUIREMENTS

The Government of Saskatchewan (GoS) through SaskBuilds and Procurement's Information Technology Division (ITD) hosts and offers multiple Enterprise Service offerings either directly through ITD or through its designated third-party service partners. **Appendix F: ITD Environment** can be referenced for a list of the current Enterprise Standards and associated Services.

For clarity, the following section 2.1 outlines the GoS Enterprise Services **that are applicable to all solutions**. These services are required to be included in all proposed solution designs. The Proponent is to describe how these requirements will be met in their proposal, including any related assumptions.

2.1 Required Enterprise Services and Standards

GoS requires Proponents to utilize, integrate or comply with the following Enterprise Services and Standards:

2.1.1 Application Authentication

Efficient management of user identities, single sign-on capabilities and overall security management are part of a One Government approach to service delivery. All solutions that are dependent on user identities and authentication are required to utilize or integrate with the existing Government of Saskatchewan application authentication standards for employees, citizens and businesses. These include:

Government Employee Identities and Authentication - Applications or functionality that are targeted for government employees must utilize the GoS Microsoft Azure Active Directory. Integration with Azure AD can be through Single Sign On (SSO) deployments such as OpenID Connect/OAuth2.0 or SAML2.0. Where applicable, Business to Business (B2B) delegation to the GoS Azure AD as the identity and authentication source is also possible.

Citizen and Business Identities and Authentication - Applications or functionality that are targeted for citizens or businesses must utilize the GoS Saskatchewan Account identities and authentication (currently provided by Vivvo CitizenOne). Integration with Saskatchewan Account can be through Single Sign On (SSO) SAML2.0 deployments.

Government Partner Organizations - Applications or functionality that are targeted for businesses providing services to the Government of Saskatchewan are required to use the GoS Azure AD Business to Business (B2B) delegation with their identity provider.

2.1.2 Enterprise Digital Standard

GoS has undertaken focused efforts to standardize how citizens and organizations conduct online transactions with government. As such:

- The Saskatchewan.ca website must be the single entry point for all content and digital services offered by the GoS. Ministries will use Saskatchewan.ca as their primary promotional web address.
- Use of Saskatchewan Account (described above) is required for any services where authentication is required for public facing services.
- The solution must leverage Saskatchewan Account to manage citizen and organization profiles. All updates to profile will be complete in Saskatchewan Account and populated in the solution.

2.1.3 Enterprise Integration Services

Solutions requiring application interfaces to any government information / data assets (regardless of where they are hosted) must leverage the GoS Enterprise Integration Services (EIS) Platform. It should be noted the EIS Platform is used to integrate public entity profiles created in Saskatchewan Account (referenced above) with service provider applications using a secured, consistent and repeatable process. The EIS Platform may also provide APIs to access partner services. The current GoS integration platform is based on Microsoft Azure Integration Services (AIS).

2.1.4 Web Application Firewall Service

As part of an overall security program to mitigate risks for cyber threats against the GoS's public facing and internal applications, ITD has subscribed to a Web Application Firewall as a Service solution. This service will be incorporated into all new public facing or internal application solutions.

2.1.5 IT Service Management

GoS has adopted the ServiceNow platform for IT Service delivery (incident, problem, change, release, and knowledge management). In support of providing a "one number to call" approach for Government employees, and services being proposed, proponents must indicate how their service management processes and systems will integrate with the GoS Enterprise IT service management operations.

2.2 Enterprise Services – Leverage Opportunities

For the purposes of reducing overall redundancies and costs of the Government's IT Portfolio, and where relevant or applicable to the solution, GoS encourages Proponents to utilize the following enterprise services. Proponents should describe how these services may be incorporated into the proposed solution, including any related assumptions.

2.2.1 Enterprise Geographic Information System (eGIS)

GoS utilizes the Esri Enterprise GIS suite of products as the standard for mapping and geospatial information. The Enterprise eGIS deployment enables the government to manage and maintain information in either standard spatial data formats – vector or raster content.

2.2.2 Enterprise Customer Relationship Management (eCRM)

GoS has adopted the Microsoft Dynamics 365 Online and the PowerApps platform as the standard for GoS Ministries in managing Government's interactions with citizens and businesses of the province.

2.2.3 Enterprise Service Management, Inquiry Management

GoS has adopted the ServiceNow platform for Enterprise Service Management. In addition to supporting IT service delivery, the solution has also been extended to provide Inquiry Management capabilities for individual Ministries.

2.2.4 Enterprise Collaboration and Document Management

GoS has adopted the Microsoft SharePoint Online and Microsoft Teams as the collaboration suite for the Ministries within Executive Government.

2.2.5 Enterprise Operational Reporting (eORS)

GoS has adopted the IBM InfoSphere and Cognos Analytics platforms to host managed reports, Ministry ad-hoc reports and data for the GoS. The services within eORS include extract/transform/load ("ETL"), data store or warehouse development, business intelligence and analytics reporting capabilities.

2.2.6 Enterprise Financial Management

GoS currently utilizes Oracle e-Business Suite as an enterprise solution.

2.2.7 Electronic Payments

GoS currently utilizes Paysafe for card not present (on-line) payments, and Moneris for point of sale transactions. Note – all applications requiring electronic payments will be through direct integration with Paysafe to maintain PCI compliance.

2.3 Considerations for Using GoS Enterprise Services

The Ministry requires the Proponent's proposed approach meet the following Enterprise Service requirements when utilizing GoS enterprise service offerings:

- The Solution should take into consideration that all of **Section 4 "GOVERNMENT OF SASKATCHEWAN COMPUTING ENVIRONMENT CONSIDERATIONS"** (below) apply for this requirement;
- The Solution should consider utilizing GoS enterprise application stack for integrations. For example, a proposed CRM solution utilizing eCRM platform should use Enterprise SharePoint Service for document repository;
- The Solution should have the ability to utilize "Out of the Box" (OOTB) deployment followed by solution configurations to address business requirements;
- The Solution Provider should take maximum effort to reduce customizing the solution and in areas where customizations are required to address business requirements, the solution provider should submit architecture and design for the customization during the first iteration of development to the enterprise service governance committee for review, approval and re-use;
- For solutions requiring custom development of code for GoS, the Application Lifecycle Management (ALM) of all related software is required to use the GoS Microsoft Azure DevOps platform;
- The Solution should have the ability to be re-used on the Enterprise platform for one or more tenants (Ministries), as a whole solution. The Solution should have the ability to accommodate best practices or sub-components from this solution to be taken out and implemented as re-usable GoS Core Services;
- The Solution should have the ability to perform on a Multi-Tenant (Ministry) platform considering change management, governance, re-use and optimized user licensing at the enterprise level;
- The Solution should accommodate the standards, processes, procedures and policies that enforce each enterprise service;
- The Solution Provider should work with GoS designated third-party service partners and with other development Proponents as needed on the enterprise platform;
- The Solution should follow industry best practices for development and deployment as mandated by the application vendor in addition to following technical standards as specified in **Appendix F, ITD Environment**.

3. CLOUD COMPUTING ENVIRONMENT CONSIDERATIONS

This section applies to proposals in which the hosting of the solution will be outside of the Government of Saskatchewan's provided computing environment (which currently includes a managed "on-premises" data center and private cloud Infrastructure as a Service provider). This section applies to Proponents providing Software as a Service (SaaS) or Managed Services solutions.

3.1 Infrastructure Hosting Facility Requirements

The Ministry requires the Proponent's proposed Solution and Services to meet the following requirements:

- Hosting facilities must be located in Canada for Class A to C data sensitivity classifications;
- Hosting facilities should incorporate levels of redundancy and resiliency to facilitate achievement of application and infrastructure availability, recovery, and other objectives specified in this RFP;
- Hosting facilities should demonstrate compliancy with CSAE 3416 (formerly CICA 5970), or equivalent, audit requirements;
- Hosting facilities should hold an Uptime Institute certification or other industry data centre recognitions for data center capabilities and performance.

3.2 Identity and Access Management Requirements

The Ministry requires the Proponent's proposed approach meet the following Identity and Access Management requirements:

- The Solution must meet the authentication requirements outlined in section 2.1.1 Application Authentication (above);
- The solution should support two (or multi) factor authentication;
- The Solution should also accommodate the Identity and Access Management policy that enforce Conditional Access and Multifactor Authentication of Azure Active Directory;
- The solution must support protocols such as Security Assertion Markup Language (SAML), Open ID and OAuth for authentication and authorization; The Solution should also accommodate the user access via Azure Application Proxy for Single Sign on capability;
- The Solution should have identity and access management capabilities based on established platforms in which multiple form factors are available rather than custom-developed approaches; and
- The Solution should have identity and access management capabilities which align with the Pan-Canadian Identity Management Framework

3.3 Networking Requirements

The Ministry requires the Proponent's proposed approach meet the following Networking requirements:

- The proponent is responsible for providing appropriate access, capacity and resiliency for network connectivity to their proposed solution;
- For access to the proposed solution by GoS employees from the GoS network, the proponent should advise of any extraordinary capacity requirements that may need to be addressed by the GoS network infrastructure providers. This includes advising of specific requirements if a private network connection between GoS and the Solution is needed;

3.4 Application and Data Integration Requirements

The Ministry requires the Proponent's proposed approach meet the following Application and Data Integration requirements.

- The Solution and Services should demonstrate mature, documented application and data integration capabilities and methodologies based on industry standard technologies;
- As outlined in section 2.1.3 Enterprise Integration Services, solutions requiring application interfaces to any government information / data assets (regardless of where they are hosted) must leverage the GoS Enterprise Integration Services (EIS) Platform.

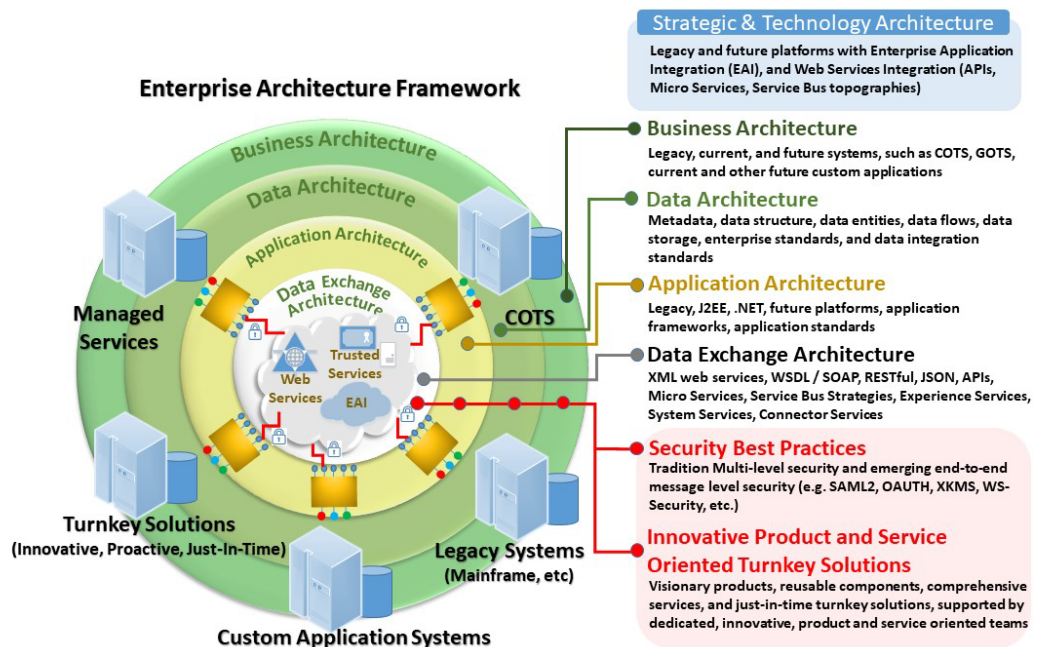


Figure: 3.4 GoS Integration Layer

- The Solution and Services should accommodate application and data integration to GoS applications utilizing the following methods:
 - Web services as part of the GoS Enterprise Integration Service (EIS)
 - Advanced integration capabilities such as Proponent delivered Enterprise Service Bus/Broker services;
 - Real-time interfaces are preferred. Where file-based transfer is necessary, secure protocols are required (Secure File Transfer Protocol - SFTP).

3.5 Data Backup, Storage, and Retention Requirements

The Ministry requires the Proponent's proposed approach meet the following Data Backup, Storage, and Retention requirements:

- The Solution and Services should incorporate established data backup, retention, and storage processes, policies, and/or practices;
- The Solution and Services should take into account that backup to removable media, if applicable, must be encrypted and securely transported and stored;
- The Solution and Services should take into account that storage locations for physical media, if applicable, shall be secured locations with limited, monitored access;
- The Solution and Services should take into account that backup of production data should facilitate restoration to any restore point in the previous 14 days;
- The Solution and Services should take into account that backup of system environments should retain a minimum of five (5) full backups from which an environment restoration can be performed;
- The Solution and Services should keep GoS data separate from the data of other clients.

3.6 End-user Computing Configuration Requirements

The Ministry requires the Proponent's proposed approach meet the following End-user Computing Configuration requirements:

- The Solution delivery to end-user browsers must support GoS active standard or supported web browsers as specified in **Appendix F, ITD Environment**;
- In cases where the Solution requires a desktop client or other similar software, the Solution should support delivery of such software on supported virtual desktops as specified in **Appendix F, ITD Environment**.

3.7 Disaster Recovery Requirements

The Ministry requires the Proponent's proposed approach meet the following Disaster Recovery requirements:

- The Solution and Services should include an established disaster recovery plan for the proposed facilities and infrastructure;
- The Solution and Services should include disaster recovery services that facilitate achievement of the disaster recovery objectives specified in this section;
- The Solution and Services should include rehearsal of the disaster recovery plan on a regular basis. The Ministry, and any other service delivery stakeholders, may need to be involved in the testing of the recovered Solution;
- In the event a disaster or disruption is declared, the Solution and Services should include recovery of the Solution in accordance with the disaster recovery plan established by the Proponent;
- In the event a disaster or disruption is declared, the Solution and Services should include operation of the disaster recovery facilities and environments to the same service levels as that mutually agreed-upon for production facilities and/or environments.

Unless otherwise noted in the RFP, at a minimum the Ministry requires the Proponent's proposed Solution and Services to take into account the following Recovery Objectives:

Recovery Time Objective (RTO):	48 hours
Recovery Point Objective (RPO):	24 hours

Proponents should indicate their base Service Levels for Recovery Time (RTO) and Recovery Point (RPO) objectives.

3.8 Availability Requirements

The Ministry requires the Proponent to recommend an adequate level of infrastructure and application monitoring and reporting to ascertain that availability objectives are being met. Unless otherwise noted in the RFP, at a minimum the Ministry requires the Proponent's proposed Solution and Services to meet the following availability objectives.

Production Availability:	99.9%, 24 hrs/day, 365 days/year
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The Ministry does not expect downtime resulting from scheduled, approved changes to be included in availability/uptime calculations.

Due to the nature of the Solution, availability exceeding this minimum requirement stated above (approaching 99.999%) could become necessary and, as such, scalability of the Solution to meet any such requirements may be necessary. Proponents are to outline options for increased levels of availability and impacts to the proposed solution.

3.9 Infrastructure/Architecture Requirements

The Ministry requires the Proponent's proposed approach meet the following Infrastructure and Architecture requirements:

- The Solution and Services should be configured, sized, and scoped in accordance with availability, recovery, service level, and other objectives specified throughout this RFP.

3.10 Other Off-premise Solution Delivery Requirements

The Ministry requires the Proponent's proposed approach to take into consideration the following Other Off-premise Solution Delivery Requirements:

- The Solution and Services should be able to accommodate scaling-up or scaling-down to meet increased or decreased demand over the lifecycle of the Solution;
- The Solution should provide mechanisms by which performance issues between customers can be prevented or isolated;
- The Solution should have established and active user groups.

3.11 Infrastructure and Hosting Services Requirements

In an off-premise solution delivery approach, the Ministry requires the Proponent to include infrastructure and hosting services that take into account the following requirements:

- The Proponent is to maintain up to date documentation of the solution, including but not limited to the architecture and configuration; change and release plans, and operational procedures. ITD is given the right to audit the documentation, and/or can be given a copy upon request;
- Where applicable, the Services should be governed by established, mature, and documented service management practices including, though not necessarily limited to, incident management, problem management, release management, change management, capacity management, configuration management, and information security management;
- The Solution and Services should include established system and application monitoring and alerting capabilities including monitoring of servers, network, storage, and any other supporting elements and components of the Solution;
- The Services should include established practices/processes for preventing/mitigating infrastructure performance issues;
- The Services should include established practices/processes for virus scanning and mitigation, vulnerability analysis and mitigation, and patch management for all supporting elements of the Solution;
- The Services should include established patching schedules that include infrastructure updates, operating system patching and updates, software platform patches and updates, and Solution patching (NOTE: It is expected at all critical and security-related patches are applied to supporting elements of the Solution in a timely manner);

- The Services should take into account that any changes to the Solution or its supporting elements shall take place within mutually agreed-upon maintenance windows;
- The Services should include proactive maintenance, updates, and upgrades that ensure all supporting elements of the Solution are maintained at a current ("n") or n-1 level;
- The Services should include design and execution of changes to supporting elements of the Solution made necessary by changes to the Solution.
- The Services should support future integration for exchanging realtime system health information (operational status, performance, patch status, etc.) via APIs with GoS Cloud Service Management operations.

3.12 Exit Strategy

In the event of contract expiry or termination, or significant changes to business or technical requirements, the Solution and Services should include an exit strategy for the system which includes, to the extent applicable, return of the system, application, and data in a format which can be recovered to another environment specified by the GoS. The Proponent is also responsible to remove all GoS data, and proof that the government information (including backups) have been removed from their environment.

4. GOVERNMENT OF SASKATCHEWAN COMPUTING ENVIRONMENT CONSIDERATIONS

The following section applies to proposals which are intended to be hosted within the existing Government provided infrastructure. The section outlines items to be considered if proposing solutions that will be dependent on Government of Saskatchewan's services associated with its network, managed on-premises data center and/or private cloud services environment.

4.1 Infrastructure Hosting Facility Requirements

The GoS requires the Proponent's proposed Solution and Services to take into account the following requirements and considerations pertaining to the Infrastructure Hosting Facility:

- The Solution and Services must take into account that the GoS datacentre is located in Saskatchewan, Canada and that all necessary infrastructure and managed infrastructure and hosting services will be provided from this location;
- The Solution and Services should take into account that the GoS datacentre is logically divided into separate VLANs between data tiers, application tiers, and public facing zone/web proxy;
- The Solution and Service should also take into account the GoS ability to host the solution within the GoS tenancy in the Azure Canada Region. The GoS environment has been designed to include vendor-managed branches of Azure Management Groups.

4.2 Identity and Access Management Requirements

The GoS requires the Proponent's proposed Solution and Services to take into account the following requirements pertaining to Identity and Access Management:

- The Solution must meet the authentication requirements outlined in section 2.1.1 Application Authentication (above);
- Solutions requiring use of Privileged Identities and Authentication (for solution support or administration) must utilize B2B and the GoS Privileged Identity Management service (Cyberark).
- The Solution must restrict access by authenticated users via the Government of Saskatchewan Azure Active Directory. Once the user is authenticated for access to the solution, the user activity within the application must be restricted by granted authorizations within Azure Active Directory or by roles/privileges within the application. The Ministry has a preference that security groups be used to enable this authorization;
- The Solution should consider the hybrid identity model of user authentication between On-Premise Active directory and Azure Active Directory;

- The Solution should also accommodate the Identity and Access Management policy that enforce Conditional Access and Multifactor Authentication of Azure Active Directory;
- The Solution should also accommodate the user access via Azure Application Proxy for Single Sign on capability;
- The Solution should accommodate roles-based authorizations and access privileges rather than user-based;
- The Solution should not allow access to application data via methods other than through the application. Ad hoc queries/reports that are not authenticated and logged are NOT permitted, even by administrative staff;
- The solution must support protocols such as Security Assertion Markup Language (SAML), Open ID and OAuth for authentication and authorization; and
- The solution should support two (or multi) factor authentication. The Government has implemented two factor authentication with the Entrust Identity Guard software.

4.3 Networking Requirements

The GoS requires the Proponent's proposed Solution and Services to take into account the following requirements pertaining to Networking:

- The Solution should accommodate deployment using load balancers and/or application level server clusters;
- The Solution should take into account that access between VLANs is restricted unless specific ports are opened for communications access between zones (e.g.: an application server that needs to store and/or read data from a database server);
- The Solution should take into account that end-user access to the Solution will be from offices from across the province of Saskatchewan. Many offices have high speed network connections; however, some offices have speeds as low as 1.5 Mbps. Furthermore, some employees will access the Solution from remote locations where there is minimal cellular coverage and internet access;

4.4 Remote Access Requirements

The GoS requires the Proponent's proposed Solution and Services to take into account the following requirements pertaining to Remote Access:

- Remote access to the environment will be accommodated by either Virtual Private Network (VPN) access using ITD's standards; or
- Privileged identity and access is administered through the Government of Saskatchewan's Identity and Access Management team utilizing remote access technologies provided through Cyberark.

4.5 Application and Data Integration Requirements

The GoS requires the Proponent's proposed Solution and Services to take into account the following requirements pertaining to Application and Data Integration:

- The Solution should ensure data integration meets minimum requirements to support integration of required data via one of the following interface methods:
 - file-based (XML, CSV);
 - inbound email connectors;
 - message-based (Web Service);
 - database to database level integration;
 - industry standard integration techniques, such as asynchronous and synchronous Application Programming Interfaces (API), JSON, XML, REST, WSDL, EDI, Enterprise Application Integration (EAI) technologies, and others.
- The Ministry has a preference towards solutions that provide message-based integration capabilities;
- The Solution should accommodate file transfers via Secure File Transfer Protocol (SFTP). In cases where a file based transfer includes files transferred outside of the GoS security perimeter, it must use web services or SFTP;
- As outlined in **section 2.1.3 Enterprise Integration Services**, solutions requiring application interfaces to any government information / data assets (regardless of where they are hosted) must leverage the GoS Enterprise Integration Services (EIS) Platform.

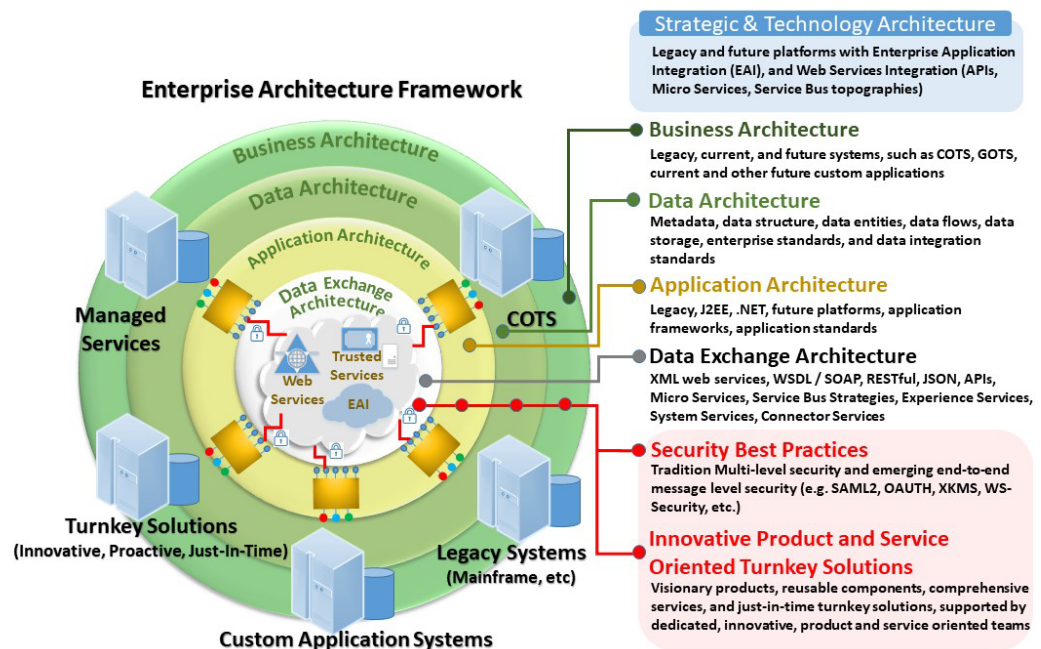


Figure: 4.5 GoS Integration Layer

4.6 Application Lifecycle Management (ALM)

For solutions requiring custom development of code for GoS, the management, configuration, testing and deployment of all software is required to use the GoS Microsoft Azure DevOps platform;

4.7 Data Backup, Storage, and Retention Requirements

The GoS requires the Proponent's proposed Solution and Services to take into account the following requirements pertaining to Data Backup, Storage, and Retention:

- The Solution should accommodate ability for file-based system outputs;
- The Solution should take into account that data backups for production data facilitates restoration to any restore point in the previous 14 days;
- The Solution should take into account that data backups for system environments retain a minimum of five (5) full backups from which an environment restoration can be performed.

4.8 End-user Computing Configuration Requirements

The Ministry requires the Proponent's proposed approach meet the following End-user Computing Configuration requirements:

- The Solution delivery to end-user browsers must support GoS active standard or supported web browsers as specified in **Appendix F, ITD Environment**;
- In cases where the Solution requires a desktop client or other similar software, the Solution should support delivery of such software on supported virtual desktops as specified in **Appendix F, ITD Environment**.

4.9 Disaster Recovery Requirements

Unless otherwise noted in the RFP, at a minimum the Ministry requires the Proponent's proposed Solution and Services to take into account the following Recovery Objectives:

Recovery Time Objective (RTO):	48 hours
Recovery Point Objective (RPO):	24 hours

Proponents should indicate their base Service Levels for Recovery Time (RTO) and Recovery Point (RPO) objectives.

The GoS requires the Proponent's proposed Solution and Services to take into account the following requirements pertaining to Disaster Recovery:

- The Solution and Services should take into account that the ITD Disaster Recovery Plan leverages, as much as possible, current GoS infrastructure including offsite backups and recovery at a cold site;
- The Solution and Services should take into account that the ITD Disaster Recovery Plan is rehearsed on an annual basis;
- The Solution and Services should take into account that the Successful Supplier will be expected to participate in disaster recovery rehearsals for recovery of the application-related components of the Solution;
- The Solution and Services should take into account that the Successful Supplier will be expected to work collaboratively with the ITD to continuously improve the Disaster Recovery Plan and documentation and processes;
- The Solution and Services should take into account that, in the event of the declaration of disruption or disaster, the Successful Supplier will be expected to deliver the Services to the Ministry under the same service level requirements.

4.10 Availability Requirements

Unless otherwise noted in the RFP, at a minimum the Ministry requires the Proponent's proposed Solution and Services to meet the following availability objectives:

Production Availability: 99.9%, 24 hrs/day, 365 days/year

The Ministry does not expect downtime resulting from scheduled, approved changes to be included in availability/uptime calculations.

Due to the nature of the Solution, availability exceeding this minimum requirement stated above (approaching 99.999%) could become necessary and, as such, scalability of the Solution to meet any such requirements may be necessary. Proponents are to outline options for increased levels of availability and impacts to the proposed solution.

4.11 Infrastructure/Architecture Requirements

The GoS requires the Proponent's to propose the infrastructure required to host the Solution within the GoS datacentre. In proposing the recommended infrastructure for the proposed Solution, Proponents should take into account the following requirements:

- To the extent possible, recommended infrastructure should align with that which is already supported in the GoS datacentre;
- Solution Infrastructure and architecture proposed by the Proponent should be adequate to meet availability, recovery, and other objectives stated throughout this RFP.

4.12 Other On-premise Solution Delivery Requirements

The Ministry requires the Proponent's proposed approach to take into consideration the following Other On-premise Solution Delivery Requirements:

- The Solution and Services should be able to accommodate scaling-up or scaling-down to meet increased or decreased demand over the lifecycle of the Solution;
- The Solution should have established and active user groups.

4.13 Infrastructure and Hosting Services Requirements

The Ministry will require the Successful Supplier to work collaboratively with ITD, or its designated third-party service partner, to ensure the operational integrity of the Solution and all supporting elements of the Solution.

4.14 Exit Strategy

In the event of contract expiry or termination, or significant changes to business or technical requirements, the Solution and Services should include an exit strategy for the system which includes, to the extent applicable, return of the system, application, and data in a format which can be recovered to another environment specified by the GoS.